

Differential Amplifiers Data Collection

Table 1: Differential Amplifier with Tail Resistor Circuit Data.

R_{Tail}	V_{CC}	V_{EE}	V_{RC}	V_{CE}	V_{RTail}
<i>Calculated</i>					
<i>Measured</i>					
R_{Tail}	$R1 \ \& \ R2$	$R3 \ \& \ R4$	$R5$	RL	
<i>Calculated</i>					
<i>Measured</i>					
R_{Tail}	$P_{R1} \ \& \ P_{R2}$	$P_{R3} \ \& \ P_{R4}$	P_{R5}	P_{RL}	$P_{Q1} \ \& \ P_{Q2}$
<i>Calculated</i>					
<i>Measured</i>					
R_{Tail}	Z_{IN}	Z_{OUT}	ΔV_{CE}	ΔV_{CC}	ΔV_{CB}
<i>Calculated</i>					
<i>Measured</i>					
R_{Tail}	ΔV_{CCCB}	ΔV_{CM}	ΔV_{Diff}	$CMRR$	$CMRR_{dB}$
<i>Calculated</i>					
<i>Measured</i>					

Table 2: Differential Amplifier with Constant Current Source Circuit Data.

<i>CCS</i>	V_{CC}	V_{EE}	V_{RC}	V_{CE}	V_{CCS}
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	$R1 \ \& \ R2$	$R3 \ \& \ R4$	$R5$	$R6$	RL
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	$P_{R1} \ \& \ P_{R2}$	$P_{R3} \ \& \ P_{R4}$	P_{R5}	P_{R6}	
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	$P_{Q1} \ \& \ P_{Q2}$	P_{Q3}	P_{Q4}		
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	Z_{IN}	Z_{OUT}	ΔV_{CE}	ΔV_{CC}	ΔV_{CB}
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	ΔV_{CCCB}	ΔV_{CM}	ΔV_{Diff}	$CMRR$	$CMRR_{dB}$
<i>Calculated</i>					
<i>Measured</i>					
<i>CCS</i>	$\Delta V -$	$\Delta V +$	$V_{out_{maxP}}$	FC_{LOW}	FC_{HIGH}
<i>Calculated</i>	-10	+10		20hz	20Khz
<i>Measured</i>					